

ORIGINAL ARTICLE

Quantitative Risk Assessment of Diisocyanates Leaching From Wound Dressings Under Physiological Conditions

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ABSTRACT

Introduction: Diisocyanates are potent sensitizers extensively used in polyurethane-based materials, with numerous case reports linking them to allergic contact dermatitis (ACD). Recent extractables studies have detected residual diisocyanates in medical devices such as wound dressings. However, these assessments typically rely on non-physiological extraction methods, limiting their relevance to actual skin exposure conditions.

Objectives: This study aimed to develop and validate a physiologically relevant method for quantifying residual 4,4'-diphenylmethane diisocyanates (MDI), dicyclohexylmethane-4,4-diisocyanates (DMDI), hexamethylene diisocyanates (HDI), toluene diisocyanates (2,4- and 2,6-TDI) and isophorone diisocyanates (IPDI) leaching from wound dressings, and subsequently assess their risk of sensitisation induction and ACD elicitation.

Methods: Diisocyanate standards were stabilised as di-lysine adducts in an artificial sweat solution, simulating skin exposure. An optimised LC–MS/MS method was used for quantification. The method was validated across a wide concentration range and subsequently applied to 36 commercial wound dressings. Exposure levels were evaluated against toxicological thresholds for quantitative risk assessments.

Results: The method proved reliable for all target diisocyanates (except IPDI) at concentrations ≥ 15 ppb. Analysis of the samples revealed that 44% contained quantifiable levels of residual diisocyanates. Notably, over 50% of the HDI-positive and 20% of MDI-positive samples exceeded the acceptable exposure level (AEL) for sensitisation induction. Moreover, all samples exceeding the AEL fell above the acceptable non-eliciting area dose for contact dermatitis (ANEAD).

Conclusion: We developed and validated a physiologically relevant method to quantify diisocyanate residues leaching from wound dressings, demonstrating that HDI, in particular, poses a significant risk for sensitisation induction and can elicit ACD in individuals who are already sensitised.

1 | Introduction

Diisocyanates are highly reactive organic compounds characterised by two electrophilic $N=C=O$ functional groups, which can readily react with nucleophiles such as water, alcohols, and amines to form, amines, urethanes, and ureas,

respectively [1]. Their main industrial application is in the production of polyurethane (PU) through reactions with polyols. PU is widely used in a wide variety of products such as foams, adhesives and sealants [2]. The most commonly used diisocyanates in PU manufacturing include toluene-2,4-diisocyanate (TDI-2,4), toluene-2,6-diisocyanate (TDI-2,6),

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hexamethylene diisocyanate (HDI), 4,4'-diphenylmethane diisocyanate (MDI), dicyclohexylmethane-4,4'-diisocyanate (DMDI) and isophorone diisocyanate (IPDI). Each of these compounds has distinct chemical characteristics that affect the physical properties and stability of the final product after reaction with a polyol [3].

Across the European Union, numerous cases of adverse health effects have been linked to occupational exposure to (di)isocyanates [4–6]. Indeed, with more than 5000 new cases reported annually, occupational asthma is the most common reported consequence [7]. However, diisocyanates are not only known for their ability to cause respiratory sensitisation; they are also recognised as potent skin sensitizers. A study from 2018 evaluated the skin sensitisation potential of diisocyanates using the local lymph node assay (LLNA) and direct peptide reactivity assay (DPRA), and classified them as extreme sensitizers [8]. These results correlate with multiple case reports of Allergic Contact Dermatitis (ACD) in workers exposed to diisocyanates [9–11].

While occupational exposure to diisocyanates has long been recognised, recent reports have raised concerns that these compounds may also cause ACD in patients using PU-containing medical devices, such as wound dressings and glucose monitoring devices [12]. Indeed, PU is widely used in biomedical and clinical applications due to its favourable biocompatibility and biostability [13]. Under the EU REACH regulation, however, restrictions on diisocyanates are solely targeted toward their safe handling. Product availability is not restricted, and regulatory requirements for consumer protection are only triggered when diisocyanate concentrations exceed 0.1% by weight [14]. Moreover, it is assumed that no diisocyanate monomers can be found in finished products as they are used up to form the PU by reacting with the polyols [2]. However, extractables studies conducted on medical devices have detected traces of unreacted diisocyanates [12, 15], challenging this assumption. These studies, though, often use aggressive solvents that do not reflect physiological conditions, potentially leading to an overestimation of real-life exposure [16]. In contrast, a previous leaching study using a sweat simulant found no detectable migration of residual diisocyanates [17]. Yet, this finding is difficult to reconcile with clinical reports of sensitisation linked to PU-containing wound dressings. Therefore, the aim of this study was to first develop and validate—as defined in Section 2.1.8—an analytical method to quantify leaching diisocyanates from wound dressings under realistic physiological conditions, and subsequently assess whether individuals are at risk of sensitisation induction and eventually developing ACD.

2 | Materials and Methods

2.1 | Analytical Method

2.1.1 | Standards and Reagents

The following reference standards were purchased from TCI chemicals (Antwerp, Belgium): 4,4'-diphenylmethane diisocyanate (MDI) (CAS: 101-68-8), dicyclohexylmethane-4,4'-diisocyanate (DMDI) (CAS: 5124-30-1), hexamethylene diisocyanate (HDI) (CAS: 822-06-0), toluene-2,4-diisocyanate (TDI) (CAS: 584-84-9), toluene-2,6-diisocyanate (TDI) (CAS:

91-08-7) and isophorone diisocyanates (IPDI) (CAS: 4098-71-9). The following reagents were purchased from Merck (Darmstadt, Germany): L-lysine (CAS: 56-87-1), NaCl (CAS: 7647-14-5), KCl (CAS: 7447-40-7). Acetone (grade HPLC) was bought from Biosolve Chimie (Valkenswaard, Netherlands), formic acid from Merck (Darmstadt, Germany) and acetonitrile from Biosolve Chimie (Valkenswaard, Netherlands).

2.1.2 | Synthesis of Reference Compounds

Due to their high reactivity, analyzing diisocyanates without prior stabilisation can be challenging. To overcome this, derivatisation using various reagents or solvents is often employed to produce stable derivatives [16, 18]. In this study, stable derivatives were formed by reacting diisocyanate standards with L-lysine present in the leaching solution (see further). Each standard was weighed and dissolved in 20 mL of acetone to achieve a concentration of 5000 ppm. Stabilisation was achieved by adding 1 mL of the diisocyanate solution dropwise into 20 mL of leaching solution under magnetic stirring, followed by overnight incubation in a water bath at 32°C. The reacted solutions were then filtered using a 0.45 µm mixed cellulose filter and diluted 1:100 with the leaching solution. A mixed solution containing all standard solutions at equal concentrations was also prepared and stabilised using the same protocol.

2.1.3 | Leaching Conditions

A leaching solution was prepared containing 1.25 g/L L-lysine, 2 g/L NaCl, and 0.3 g/L KCl in MilliQ water. This solution mimics skin conditions and stabilises the highly reactive diisocyanates. Indeed, in aqueous solutions with functional groups containing active hydrogens, diisocyanates preferentially react with more reactive nucleophiles than water, such as N-centered and S-centered nucleophiles found in skin proteins [19]. Previous studies have demonstrated the formation of adducts between diisocyanates and proteins [20–22], particularly identifying diisocyanate-induced modifications of keratins in human skin, where stable lysine mono- and di-adducts were formed [23]. That research informed the determination of lysine concentration in the leaching solution. By considering that 100 mg of human callus was suspended in 200 µL of DMSO, and assuming keratins consist of up to 25% nucleophilic acids that can interact with diisocyanates to form haptens, an excess lysine concentration of 1.25 g/L was chosen [24]. The salt concentrations reflect those typically found in eccrine sweat [25, 26].

2.1.4 | LC-MS/MS Conditions

The analyses were performed using an Acquity UPLC system (Waters, Milford, USA) coupled with a Xevo TQ-S triple quadrupole mass spectrometer (MS, Waters, Milford, USA). The mobile phases consisted of water with 0.2% (v/v) formic acid (mobile phase A) and acetonitrile with 0.2% (v/v) formic acid (mobile phase B). The weak and strong washes were freshly made before each sequence of injections, and were composed of 0.2% (v/v) formic acid in 10% (v/v) acetonitrile: water and 0.2% (v/v) formic acid in 40% (v/v) isopropanol: acetonitrile,

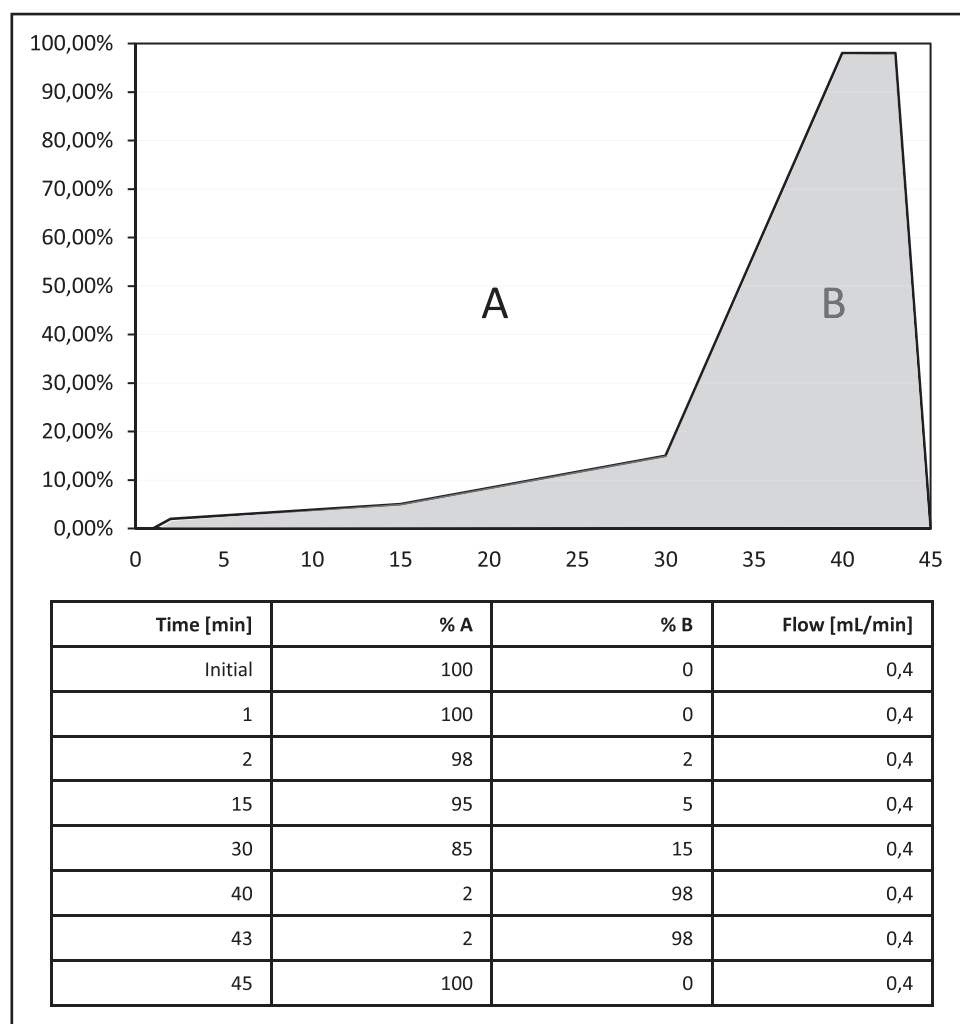


FIGURE 1 | Liquid chromatography separation gradient. The optimised gradient followed: 100% A (v/v) to 85% A in 30 min. Additionally, the gradient was lengthened by a wash step, going from 85% A to 98% B in 10 min, then maintaining 98% B for a total of 3 min, before returning to the initial condition of 100% A. The total run time was 45 min per injection. The flow delivered by the chromatographic pump was 0.4 mL/min. The mobile phases compositions were the following: Water with 0.2% (v/v) formic acid (mobile phase A) and acetonitrile with 0.2% (v/v) formic acid (mobile phase B). Min: Minutes.

respectively. The latter is crucial to avoid carry-over effects. The flow rate was set at 0.4 mL/min, and 10 µL of sample was injected into the system. Chromatographic separation was achieved on a HSS T3 column kept at a constant temperature of 45°C. A gradient, as outlined in Figure 1, was applied to optimise separation.

The electrospray ionisation (ESI) interface between the UHPLC and MS was operated in positive ion mode. Source parameters were set as follows: cone voltage of 50 V, capillary voltage of 3.0 kV, cone gas flow of 150 L/h, desolvation gas flow of 1100 L/h, desolvation temperature of 550°C, and collision gas flow of 0.25 mL/min. During the initial step of method development, Waters Intellistart was used to automatically optimise the LC-MS/MS parameters, including cone voltage, collision energy, and target-specific transitions for the lysine di-adducts of each diisocyanate under investigation. The most abundant transition was selected for quantification, while the second transition was used for ratio-based identification with an acceptance tolerance of 30%. Detailed parameters are provided in Table 1.

2.1.5 | Preparation of Validation Samples

Various wound dressings (see further, market samples) were initially scanned, and a blank sample, weighing 0.5 g, was selected for method development and validation. For validation samples of higher concentrations, each diisocyanate standard was weighed and dissolved in 20 mL of acetone to achieve a concentration of 4000 ppm. The standard solutions were mixed and diluted to reach a target concentration of 20 ppm per diisocyanate. Then, 20 mL of lysine solution was transferred into glass tubes and heated to 32°C in a water bath to simulate skin temperature. The blank samples were directly placed into the tubes and spiked dropwise with 80, 115 and 150 µL of the 20 ppm solution to reach 'on sample' concentrations of 3.3, 4.79 and 6.25 ppm, respectively. For validation samples of lower concentrations, standards were similarly dissolved at 5000 ppm in acetone, mixed, and diluted to reach final concentrations of 0.5, 1.5 and 3 ppm. A 20 mL lysine solution was again used for extraction, and dressings were dosed with 100 µL of the respective solutions to reach

TABLE 1 | Optimised LC–MS/MS parameters for Diisocyanate analysis: Parameters obtained through IntelliStart optimization for each diisocyanate analysed, including precursor and product ions, and collision energy, ensuring optimal detection and quantification of each compound.

Compound	Precursor ion (<i>m/z</i>)	Qualifier ion (<i>m/z</i>)	Quantifier ion (<i>m/z</i>)	Collision energy (V)	Retention time (min)
Toluene-2,6-diisocyanate (TDI-2,6)	234.29	84.04	173.14	20/12	4.3
Hexamethylene diisocyanate (HDI)	231.27	84.04	173.10	22/10	9.0
Toluene-2,4-diisocyanate (TDI-2,4)	234.20	84.03	173.13	22/10	7.7
Isophorone diisocyanate (IPDI)	258.33	84.03	147.12	24/12	22.2
4,4'-Diphenylmethane diisocyanate (MDI)	272.35	84.04	173.16	15	28.2
Dicyclohexylmethane-4,4-diisocyanate (DMDI)	278.39	84.04	173.15	28/10	29.4

'on sample' concentrations of 100, 300 and 600 ppb. Each concentration level was tested in triplicate over three separate days. After a 24 h incubation, extracts were sequentially filtered through 0.45 and 0.22 μm mixed-cellulose filters. For the low concentrations (100–300–600 ppb), 800 μL aliquots were transferred to vials in duplicate. For the high concentrations (3.3–4.79–6.25 ppm), 1 mL aliquots were transferred into vials in triplicate. Procedural blanks were prepared using the same protocol, but without adding a wound dressing.

2.1.6 | Standard Addition Method

In certain cases, chemical analyses may be affected by physical and/or chemical interferences originating from the sample itself, commonly referred to as matrix effects [27]. To correct for such interferences, the standard addition method (SAM) is often employed. This approach involves spiking sample aliquots with increasing amounts of pure standard, allowing for the construction of a linear calibration curve to accurately determine the amount of analyte in the sample [28, 29]. Standard addition solutions (SAS) were prepared following the same stabilisation protocol described previously for the diisocyanate standards. After stabilisation in 20 mL of lysine solution, the final target concentrations of the mixed SAS were: 10, 1 and 2 ppm. For the lower concentration validation levels (100–300–600 ppb), SAM was performed by spiking 200 μL of lysine solution into the first vial, and 200 μL of the 10 ppb SAS into the second vial. To assess the linearity of the method, an additional intermediate standard addition point was introduced between these two concentrations on the first day of validation. This procedure was consistently followed for all replicates across all concentrations. For higher concentrations (3.3–4.79–6.25 ppm), SAM was conducted by spiking 100 μL of lysine solution into the first vial, 100 μL of the 1 ppm SAS into the second vial, and 100 μL of the 2 ppm SAS into the third vial.

2.1.7 | Data Processing

Chromatograms obtained for each diisocyanate lysine-diadduct were integrated by means of MassLynx. The intensities

of each peak were then plotted against the respective concentrations spiked for the SAM. The resulting standard addition curves were used as calibration lines to determine the sample concentrations. The calculated concentrations of the validation samples were then used for method validation (see further).

2.1.8 | Method Validation

The method was validated in accordance with the Association of Official Analytical Collaboration (AOAC) guidelines for standard method performance requirements [30]. These guidelines outline the performance criteria recommended for different types of analytical methods. For quantitative methods, validation was based on several factors, including the applicability range, bias, precision, recovery, limit of quantification, and the relative standard deviation (RSD) for reproducibility (RSDr).

2.2 | Market Samples

The developed methodology was used to analyse commercially available wound dressings. Sample selection was guided by reported cases of allergic and irritant Contact Dermatitis linked to such products. In total, 36 wound dressings were purchased from the European retail market, encompassing a variety of dressing types: 5 traditional wound dressings, 15 foam dressings, 12 hydrocolloids, and 4 hydrogels. The sample preparation consisted of cutting and weighing pieces of 0.5 g of each market sample, and following the extraction procedure as described previously for the validation samples. After the extraction steps, the SAM was executed as explained in 2.1.6 and samples were injected into the LC–MS/MS system.

2.3 | Quantitative Risk Assessment (QRA) for Skin Sensitisation Induction

Exposure-based risk assessment for skin sensitisation induction was conducted following the QRA methodology, originally developed for fragrance substances, which can be summarised in four key steps [31].

2.3.1 | Hazard Identification

This step determines whether a chemical compound has the potential to cause skin sensitisation. It typically relies on experimental data from assays such as the LLNA, which measures the proliferation of lymphocytes in the draining lymph nodes following topical application of the test substance. A stimulation index (SI) of three or more indicates a positive sensitisation response. The sensitising potency of a chemical can be expressed by the Effective Concentration (EC) 3 value, which is the concentration required to elicit a three-fold increase in lymphocyte proliferation compared to controls [32]. Chemicals with EC3 values below 2% are considered strong sensitizers and EC3 values lower than 0.2% are considered extreme sensitizers [33].

2.3.2 | Hazard Quantification

In this phase, the 'no expected sensitisation induction level (NESIL)' is established. The NESIL represents the exposure level below which sensitisation is not expected to occur. The NESIL ($\mu\text{g}/\text{cm}^2$) can be derived from the EC3 values, expressed in %, using the standard dose volume used in the LLNA that is 25 μL of substance per application to a 1 cm^2 area, and assuming a density of 1 g/mL [34]:

$$\frac{\text{EC3}(\%) \cdot 25000(\mu\text{g})}{100 \cdot 1(\text{cm}^2)} = \text{NESIL} \frac{(\mu\text{g})}{(\text{cm}^2)}$$

Sensitisation assessment factors (SAFs) are then applied to account for uncertainties and differences between experimental conditions and real life conditions. These factors cover aspects such as intraspecies variability, product composition, frequency and duration of use, and skin condition at the site of exposure [35].

2.3.3 | Exposure Assessment

This step estimates the Consumer Exposure Levels (CEL) by evaluating how much of the chemical individuals are exposed to under typical usage conditions. In this study, the stabilisation of diisocyanates with lysine mimics was employed to mimic their reaction with skin proteins under physiological conditions. Consequently, the concentrations obtained from the leaching studies were considered representative of actual CEL values. By taking into account the surface area of the sample (S), the measured concentration (C), and the weight of the sample (w), the final exposure values were expressed in $\mu\text{g}/\text{cm}^2$:

$$\frac{\frac{C(\text{ppb})}{1000} \cdot w(\text{g})}{S(\text{cm}^2)} = \text{CEL} \frac{(\mu\text{g})}{(\text{cm}^2)}$$

2.3.4 | Risk Characterisation

The first step in risk characterisation involves determining the Acceptable Exposure Level (AEL), which is calculated by dividing the NESIL by the total SAF [36]. The AEL represents the

exposure threshold below which sensitisation is not expected to occur. To assess the risk, the AEL is compared to the CEL:

- If $\text{AEL} \geq \text{CEL}$: the risk of skin sensitisation induction is considered acceptable, indicating likely safe use
- If $\text{AEL} < \text{CEL}$: there is a potential risk of inducing skin sensitisation

2.4 | Quantitative Risk Assessment (QRA) for Allergy Elicitation

To protect individuals that are already sensitised to a substance from triggering an allergic reaction, an acceptable non-eliciting area dose (ANEAD), rather than an AEL, needs to be determined to ensure that exposure levels remain below the threshold capable of inducing elicitation. The ANEAD is derived using the NESIL, adjusted by an elicitation factor (EF) that reflects the sensitisation potency of the allergen and elicitation assessment factors (EAFs) that cover the same aspects as the SAFs [37]:

$$\text{ANEAD} = \frac{\text{NESIL}}{\text{EAF} \times \text{EF}}$$

With EF calculated according to Griem et al. using the log of NESIL in $\mu\text{mol}/\text{cm}^2$:

$$\text{EF} = 10^{0.84 \log\left(\text{NESIL} \left(\frac{\mu\text{mol}}{\text{cm}^2}\right)\right) + 1.81}$$

To assess the risk of elicitation, the ANEAD is compared to the CEL:

- If $\text{ANEAD} \geq \text{CEL}$: the risk of elicitation is considered acceptable, indicating likely safe use in already sensitised individuals
- If $\text{ANEAD} < \text{CEL}$: there is a potential risk of triggering an allergic reaction in individuals who are already sensitised to the allergen

3 | Results

3.1 | Method Development and Optimisation

The optimised multiple reaction monitoring (MRM) parameters are presented in Table 1. The observed transitions at m/z 173 and m/z 84 correspond to the [Lysine-NCO]⁺ and [C5H10N]⁺ fragments, respectively, aligning with the expected fragmentation patterns [23]. Notably the m/z 173 transition was not detected for the lysine di-adduct of IPDI, suggesting a distinct fragmentation behaviour for this compound.

To evaluate the specificity of the developed method, lysine-stabilised diisocyanate standards were individually injected onto the UPLC HSS T3 column. For most compounds, chromatograms revealed a characteristic pattern of three peaks with decreasing intensities and unique retention times that were observed. The three peaks are likely due to the presence of two primary amino groups on the lysine, one on the side chain and the other on the backbone, that can each react with both NCO

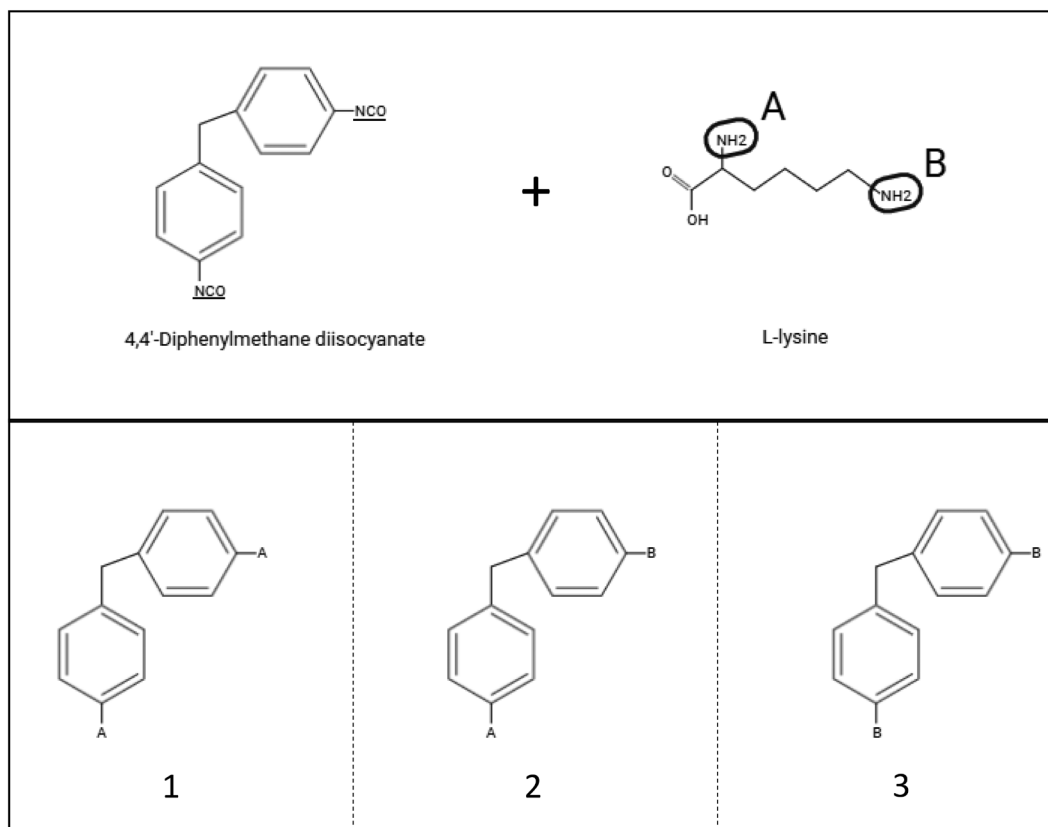


FIGURE 2 | Isocyanate functional groups and lysine primary amino groups reaction scenarios. This is an example that illustrates the various reaction scenarios between diisocyanates and L-lysine. The molecule on the right, 4,4'-diphenylmethane diisocyanate (MDI), presents 2 highly reactive isocyanate groups. The L-lysine presents 2 primary amino groups, one on the side chain (A) and the other on the backbone chain (B). The various scenarios explain the 3 distinct peaks observed for each compound as both isocyanate groups can react with both amino groups. Scenario 1: Both isocyanate groups react with the amino group on the side chain of the L-lysine. Scenario 2: Each isocyanate group reacts with one of the amino groups of the L-lysine. Scenario 3: Both isocyanate groups react with the amino group on the backbone chain of the L-lysine. NCO: Isocyanate group.

functional groups of the diisocyanates. Figure 2 illustrates these possible reactions. In contrast, the chromatograms for IPDI and DMDI exhibited multiple overlapping peaks and peak splitting, due to the isomeric forms within these standards. For identification and quantification, only the peaks with the highest intensities were considered. Figure 3 presents the final chromatograms for each diisocyanate compound, highlighting the specific peaks selected for quantification.

3.2 | Method Validation

To evaluate the linearity of the method between analyte concentration and the resulting signal, standard addition curves were constructed for all concentration levels. The resulting correlation coefficients and associated quality parameters, presented in Supporting Information S1, demonstrate linear relationships for all compounds meaning that the method is suitable for accurate quantification across the specified concentration range. Furthermore, the limits of quantification (LOQs) for each diisocyanate are presented in Supporting Information S1, calculated as $(10\sigma Y)/m$ using the standard deviation of the response method, where σY is the standard deviation of the responses and m the slope of the standard addition curves.

Based on the concentrations of the validation samples determined via standard addition curves, parameters such as relative bias, repeatability, intermediate precision and recovery were evaluated. The findings are summarised in Table 2. It is important to note that the concentrations shown in the table correspond to the levels after stabilisation in the leaching solution, not the initial sample concentrations.

Repeatability and intermediate precision are both measures of an analytical method's precision, reflecting the consistency between repeated measurements. However, they assess different aspects of precision and are evaluated under different conditions [38]. Repeatability evaluates the precision under identical conditions, and in this study refers to intra-day precision. Intermediate precision assesses the precision of the measurements under slightly varying conditions, or, in this case, across multiple days. Both parameters are expressed as RSD. According to the AOAC guidelines, acceptable repeatability (RSD_r) and intermediate precision (RSD_R) values depend on the analyte concentration (C , expressed as a mass fraction) and can be estimated using the following equations [30]:

$$\text{RSD}_r (\%) = C^{-0.15}$$

$$\text{RSD}_R (\%) = 2C^{-0.15}$$

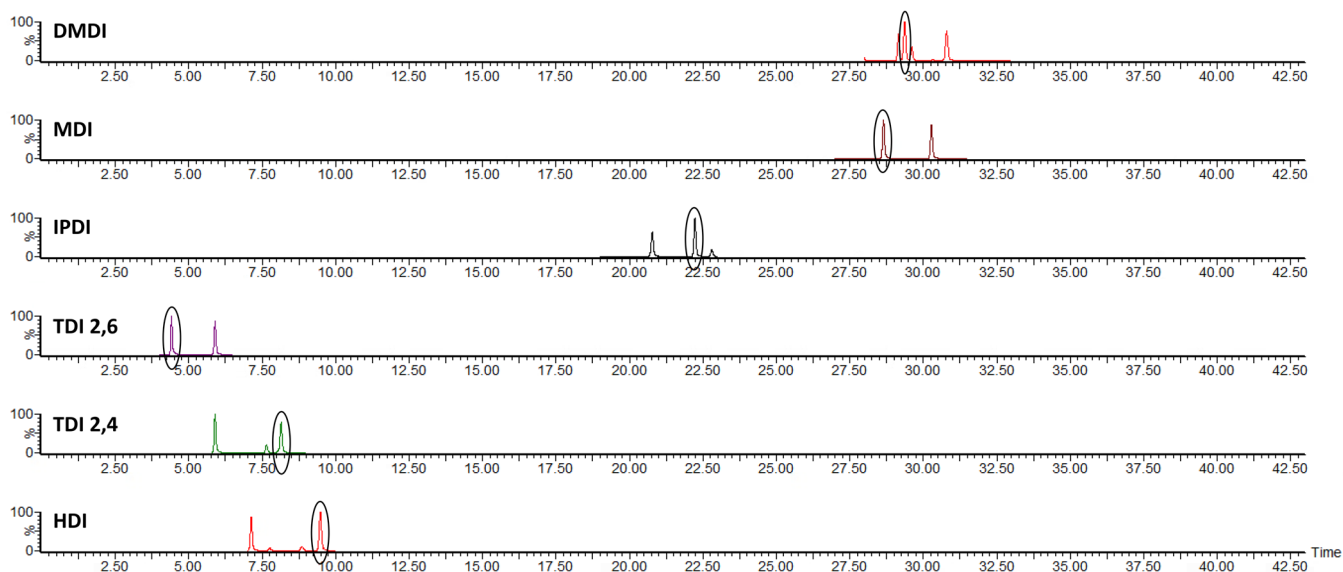


FIGURE 3 | Chromatograms obtained for each diisocyanate lysine-di-adduct studied following method optimization. As each compound exhibited three distinct peaks, the peak used for identification and quantification has been circled. DMDI, Dicyclohexylmethane-4,4-diisocyanate; HDI, Hexamethylene diisocyanate; IPDI, Isophorone diisocyanate; MDI, 4,4'-diphenylmethane diisocyanate; TDI 2,4, Toluene-2,4-diisocyanate; TDI 2,6, Toluene-2,6-diisocyanate.

The validation limits are indicated at the bottom of each concentration row in Table 2. As all RSD values presented fall within the calculated acceptable thresholds, both repeatability and intermediate precision are considered acceptable.

The recovery rate, defined as the ratio of the measured concentration to the theoretical concentration of the validation samples, serves as a critical indicator of a method's accuracy. As the concentration of the analyte decreases, the acceptable range for recovery values widens to account for increased variability inherent at lower concentrations. According to the AOAC guidelines, the acceptable mean recovery ranges for concentration units of 1, 10, 100 and 1 ppm are: 40%–120%, 60%–115%, 80%–110% and 80%–110%, respectively [39]. The results presented in Table 2 show that all recovery rates fall within these specified ranges, confirming the method's accuracy across the tested concentration levels.

The relative bias represents the trueness of an analytical method and refers to how close all measurements are to the true value that is being measured [40]. It is calculated as the percentage difference between the average measured concentration and the known reference concentration. In this study, an acceptance limit of $\pm 30\%$ was established, considering the low concentrations of the validation samples. For concentrations above 80 ppb all diisocyanates demonstrated acceptable relative bias values, with the highest observed being 5.69% for TDI 2,6, as presented in Table 2. However, at concentrations of 15 ppb and below, increased relative bias values were observed, indicating a trend of underestimation of the actual concentration that is being measured. These consistent large negative biases at the lowest concentrations may indicate inherent limitations of the method's accuracy in this range, suggesting that the method may not reliably quantify analytes at these very low levels. Despite this, the method exhibited acceptable performance across the entire concentration range

for HDI and DMDI. For TDI 2,6, the method's precision was validated starting from 7.5 ppb. Overall, the method can be considered validated for all targeted diisocyanate compounds at concentrations of 15 ppb and above. IPDI was excluded from lower concentration assessments due to its absence in initial screening samples.

3.3 | Analysis of Marketed Wound Dressings

Following method validation, the technique was applied to analyse 36 commercially available wound dressing samples. The findings, summarised in Table 3, revealed that 16 out of 36 samples contained detectable levels of one or more diisocyanates. MDI and HDI were the most frequently identified, each present in 15 samples. DMDI was detected in 14 samples, while IPDI was not detected in any of the samples. TDI isomers, specifically TDI 2,6 and TDI 2,4, were found less frequently, being present in 6 and 7 samples, respectively, typically co-occurring, with the exception of sample 7. This co-occurrence of TDI isomers aligns with industrial practices of using TDI isomer mixtures in PU manufacturing [41]. Concentration levels varied among the diisocyanates. HDI generally exhibited the highest concentrations, with 500 ppb, and one sample approaching 2 ppm. MDI concentrations surpassed 500 ppb, with 2 of them exceeding 1.5 ppm. In contrast, TDI and DMDI were present at substantially lower concentrations, with only 1 sample showing TDI 2,4 levels above 100 ppb. Notably, none of the wound dressing samples disclosed the presence of these residual monomers or provided warnings on their packaging regarding potential risks associated with their exposure.

It is noteworthy that HDI, being the most commonly detected diisocyanate at elevated concentrations in this study, is commonly used in the synthesis of 'medical-grade' polyurethanes, especially. This preference persists despite the availability of

TABLE 2 | Validation parameters for Diisocyanate analysis.

	Relative bias (%)					Repeatability (RDS %)					Intermediate precision (RDS %)					Recovery (%)				
	0.0025 (ppm)	0.0075 (ppm)	0.015 (ppm)	0.08 (ppm)	0.115 (ppm)	0.15 (ppm)	0.0025 (ppm)	0.0075 (ppm)	0.015 (ppm)	0.08 (ppm)	0.115 (ppm)	0.15 (ppm)	0.0025 (ppm)	0.0075 (ppm)	0.015 (ppm)	0.08 (ppm)	0.115 (ppm)	0.15 (ppm)	0.08 (ppm)	0.115 (ppm)
	0.0025 (ppm)	0.0075 (ppm)	0.015 (ppm)	0.08 (ppm)	0.115 (ppm)	0.15 (ppm)	0.0025 (ppm)	0.0075 (ppm)	0.015 (ppm)	0.08 (ppm)	0.115 (ppm)	0.15 (ppm)	0.0025 (ppm)	0.0075 (ppm)	0.015 (ppm)	0.08 (ppm)	0.115 (ppm)	0.15 (ppm)	0.08 (ppm)	0.115 (ppm)
TDI 2,6	-40.04	-26.32	-15.3	4.44	4.37	5.69	4.26	3.75	5.50	3.4	5.65	5.12	8.1	5.86	15.32	8.94	9.7	7.71	104.44	104.37
TDI 2,4	-56.84	-37.55	-13.44	-1.15	-1.53	-1.65	5.49	2.18	6.39	5.52	6.97	3.71	14.35	10.70	16.31	9.61	10.97	9.80	98.85	98.47
HDI	-14.31	-22.36	-14.12	-1.08	-0.66	-0.92	9.82	3.06	9.06	5.58	7.74	3.50	12.03	11.93	18.57	7.60	9.61	4.93	98.92	99.34
IPDI				3.63	3.03	1.55				3.85	4.46	4.55				8.60	9.68	8.75	103.63	103.03
MDI	-40.27	-31.63	-16.80	5.38	2.36	3.25	6.17	4.5	4.25	8.53	7.85	4.4	6.17	4.9	4.25	8.53	10.61	8.78	105.38	102.36
DMDI	-8.07	-19.36	-20.31	5.09	5.37	0.77	7.29	8.07	6.99	4.98	8.52	7.35	7.29	8.7	6.99	8.58	9.46	8.91	105.09	105.37
Limit values			(-30%; 30%)				19.51	16.55	14.91	11.60	10.99	10.56	39.02	33.10	29.83	23.20	21.97	21.12	60%-115%	80%-110%

Note: Based on the concentrations of validation samples calculated using the standard addition curves, parameters such as relative bias, repeatability, intermediate precision, and recovery were determined. Validation limits that must be met for the method to be considered validated are displayed at the bottom of the table. These values are obtained from the AOAC guidelines.
Abbreviations: DMADI, Dicyclohexylmethane-4,4-diisocyanate; HDI, Hexamethylene diisocyanate; IPDI, Isophorone diisocyanate; MDI, 4,4'-diphenylmethane diisocyanate; TDI 2,4, Toluene-2,4-diisocyanate; TDI 2,6, Toluene-2,6-diisocyanate.

less sensitising alternatives. HDI-based polyurethanes exhibit a lower hydrogen bonding index compared to those synthesised with other diisocyanates, which contributes to their superior hardness and flexibility—key properties for wound dressings [42]. In contrast, MDI is frequently present due to its influences on the permeability of wound dressings, attributed to the hydrophobic nature of MDI-based polyurethane. TDI, on the other hand, is primarily employed in the production of flexible polyurethane foams [43]. Supporting Information S2 presents all samples containing both TDI isomers. As expected, all these samples were composed of a thick, flexible foam, distinguishing them from other positive samples. These findings align with previous research on the detection of unreacted TDIs in foams [44]. Additionally, all foam samples specifically mentioned ‘polyurethane’ in their instructions for use.

3.4 | Risk Assessment

The calculated AELs and ANEADs for each investigated diisocyanate are shown in Table 4. EC3 values were retrieved from a previous study by Ohtake et al. with the exception of TDI 2,6 for which no data could be found [8]. All assessed diisocyanates were found to have EC3 values below the 0.2% threshold, classifying them as extreme sensitizers. Among the compounds studied, HDI demonstrated the highest sensitisation potency, with an EC3 value as low as 0.05%.

Currently, no official SAF values have been established specifically for wound dressings. However, the Scientific Committee on Consumer Safety (SCCS) generally recommends an interspecies SAF of 3 to address potential underestimation of human sensitisation thresholds [45, 46]. Additionally, to account for inter-individual variability, a default SAF of 10 was applied. For the product formulation factor, a SAF of 1 was used, as no permeation enhancers are expected in the matrix of wound dressings. Similarly, a SAF of 1 was assigned for frequency and duration, given the limited and controlled use of these products over extended periods. In contrast, for skin condition, a SAF of 10 was deemed necessary, as wound dressings are applied to injured and inflamed skin, which is more prone to sensitisation [47, 48]. Collectively, these values yield a total SAF of 300. The AEL for each diisocyanate (except TDI 2,6) was then calculated by dividing the NESIL by this total SAF. For the calculation of the ANEAD, the same total assessment factor of 300 was used for elicitation (EAF), with a supplementary elicitation factor (EF) derived from the correlation between the ratio of sensitisation/elicitation threshold and the sensitisation threshold taken into account. The ANEAD was then calculated using the formula from Section 2.4 above. As expected, the ANEAD values are lower than the AEL for sensitisation [49].

Table 5 presents the CELs for measured diisocyanates in the market samples, normalised to the surface area of each wound dressing. These values were used as representative CELs for risk assessment. Indeed, as wound dressings are used on damaged skin, where the barrier function is diminished and absorption is enhanced, it is assumed that the diisocyanates reacting with lysine in the leaching solution are indicative of the fraction absorbed through the skin. In Table 6, values

TABLE 3 | Concentrations of leachable, non-reacted diisocyanate residues measured in 16 out of 36 marketed wound dressings.

Sample	TDI 2,6	HDI	TDI 2,4	IPDI	DMDI	MDI
Sample 1		74.56			3.36	451.54
Sample 2	55.98	510.88	38.19		3.35	1792.85
Sample 3	99.67	257.76	70.95		24.55	51.5
Sample 4	78.11	860.67	74.07		44.79	58.74
Sample 5		1854.9			9.55	88.37
Sample 6		175.35			30.84	49.63
Sample 7		279.03	36.69		11.16	1657.43
Sample 8		92.52			18.8	119.49
Sample 9					4.49	5.21
Sample 10	68.98	704.78	59.55		82.28	657.27
Sample 11		132.96			5.04	250.83
Sample 12		523.83			22.82	9.72
Sample 13	32.78	65.53	64.84		10.41	13.54
Sample 14	52.31	31.13	103.49			9.5
Sample 15		70.07			16.81	45.01
Sample 16		23.1				

Note: Results are expressed in parts per billion (ppb) (w:w). Values shown in bold fall within the validation range of the quantification method, indicating reliable quantification. Non-bold values are below the validation range.

Abbreviations: DMDI, Dicyclohexylmethane-4,4'-diisocyanate; HDI, Hexamethylene diisocyanate; IPDI, Isophorone diisocyanate; MDI, 4,4'-diphenylmethane diisocyanate; TDI 2,4, Toluene-2,4-diisocyanate; TDI 2,6, Toluene-2,6-diisocyanate.

highlighted in bold indicate cases where the CEL exceeded the previously determined AEL and ANEAD. While the CELs for both TDI isomers and DMDI remained below the respective AEL, exceedances were observed for HDI and MDI. Specifically, more than 50% of the samples containing HDI exhibited CEL values surpassing the AEL of $0.0042 \mu\text{g}/\text{cm}^2$, with sample 10 reaching a level 10 times higher and sample 4 reaching a level 15 times higher. For MDI, three samples demonstrated CEL values exceeding the AEL of $0.0167 \mu\text{g}/\text{cm}^2$. Figure 4 shows a graphical representation of the worst cases of CELs observed for each allergen and how these values fall within the range of skin sensitisation induction risk characterisation. Moreover, all samples that had CELs above the AELs also show CELs exceeding the ANEADs. Only sample 1 and sample 4, whose CEL values were considered safe for sensitisation, have CELs for MDI and TDI-2,4 above the determined ANEAD.

4 | Discussion

The results of this study align with previous research demonstrating the presence of residual diisocyanates in medical devices, thereby challenging the assumption—often inferred from REACH regulations—that unreacted diisocyanates are absent in final consumer products [2, 12, 15]. However, while earlier studies primarily focused on extractable substances under exaggerated conditions, this investigation advances the field by quantifying the amount of diisocyanates leaching from wound dressings under physiologically relevant conditions and

potentially reacting with skin proteins (here simulated using lysine), thereby contributing to sensitisation.

The quantitative risk assessment revealed that more than 50% of the wound dressing samples containing HDI posed a potential risk for both sensitisation induction and elicitation. This finding is particularly concerning given the intended use of the analyzed samples. Wound dressings are applied to compromised skin following injuries, such as surgical procedures or physical trauma, in which harmful external objects penetrate the body. The body's normal immune response to a breach in the skin involves a controlled inflammatory reaction that initiates the healing process. Inflammation represents the second critical phase of optimal wound healing [51]. Several studies, based on both experimental and clinical data, have concluded that the presence of inflammation can increase the risk of contact sensitisation [48]. Moreover, given that elicitation generally occurs at lower doses than sensitisation, individuals that are already sensitised to HDI may be at heightened risk of developing ACD.

This concern is further amplified by the well-documented cross-reactivity among diisocyanates. A previous study in animals demonstrated that over 70% of HDI-sensitised subjects exhibited cross-reactivity to lower doses of DMDI and MDI, while more than 20% reacted to TDI. Similarly, MDI-sensitised animals showed cross-reactions to HDI (60%), DMDI (> 70%), and TDI (> 35%) [52]. Additionally, clinical cases have also reported cross-reactions, such as TDI sensitivity following exposure to MDI-containing medical devices [12]. These findings

TABLE 4 | Determination of the AEL (acceptable exposure level) and the ANEAD (acceptable non-eliciting area dose), including their EC3 values all obtained from a previous study by Ohtake et al. [8], and their NESIL (no expected sensitisation induction level).

	SAFs/EAFs					Induction		Elicitation			
	EC3 (%)	NESIL (μg/cm ²)	MW (g/mol)	Interspecies	Intraspecies	Product composition	Frequency/duration	Site condition	AEL (μg/cm ²)	EF	ANEAD (μg/cm ²)
TDI 2,6	NA	NA	174.16	3	10	1	1	10	NA	NA	NA
TDI 2,4	0.02	5	174.16						0.0167	3.27	0.005
HDI	0.005	1.25	168.19						0.0042	0.99	0.004
IPDI	0.01	2.5	222.28						0.0083	1.46	0.006
MDI	0.02	5	250.25						0.0167	2.48	0.007
DMDI	0.06	15	262.35						0.0500	5.82	0.009
Total SAF/EAF											
300											

Note: The SAFs (sensitisation assessment factors) and EAFs (elicitation assessment factors) account for uncertainties related to interspecies differences, intraspecies variability, product composition, exposure frequency/duration, and site conditions, yielding a total SAF/EAF of 300 as indicated at the bottom of the table. Abbreviations: DMDI, Dicyclohexylmethane-4,4'-diisocyanate; EF, Elicitation factor; HDI, Hexamethylene diisocyanate; IPDI, Isophorone diisocyanate; MDI, 4,4'-diphenylmethane diisocyanate; MW, Molecular weight; NA, Non-attributed; TDI 2,4, Toluene-2,4-diisocyanate; TDI 2,6, Toluene-2,6-diisocyanate.

TABLE 5 | Estimated consumer exposure levels (CEL) of wound dressing market samples.

Sample	TDI 2,6	HDI	TDI 2,4	DMDI	MDI
Sample 1		0.0027		0.0001	0.0163
Sample 2	0.0026	0.0241	0.0018	0.0014	0.0846
Sample 3	0.0031	0.0079	0.0022	0.0008	0.0016
Sample 4	0.0058	0.0637	0.0055	0.0033	0.0043
Sample 5		0.0181		0.0001	0.0009
Sample 6		0.0154		0.0027	0.0044
Sample 7		0.0082	0.0011	0.0003	0.0485
Sample 8		0.0048		0.0010	0.0062
Sample 9				0.0002	0.0002
Sample 10	0.0045	0.0462	0.0039	0.0054	0.0431
Sample 11		0.0012		0.0000	0.0023
Sample 12		0.0098		0.0004	0.0002
Sample 13	0.0023	0.0047	0.0047	0.0006	0.0008
Sample 14	0.0010	0.0006	0.0021		0.0001
Sample 15		0.0033		0.0006	0.0020
Sample 16		0.0023			

Note: Results are expressed in micrograms per square centimetre ($\mu\text{g}/\text{cm}^2$). Abbreviations: DMDI, Dicyclohexylmethane-4,4'-diisocyanate; HDI, Hexamethylene diisocyanate; MDI, 4,4'-diphenylmethane diisocyanate; TDI 2,4, Toluene-2,4-diisocyanate; TDI 2,6, Toluene-2,6-diisocyanate.

indicate that wound dressings may not only induce sensitisation to the diisocyanates they contain, but also promote cross-sensitisation, potentially leading to ACD upon exposure to a broad range of diisocyanates. This underscores the importance of rigorous evaluation of sensitisation risks associated with such products.

The newly developed analytical method was fully validated according to AOAC guidelines for all target diisocyanate compounds at concentrations ≥ 15 ppb, except for IPDI, which was excluded due to non-detection in initial screenings. The method demonstrated consistent precision across the analytical range, as evidenced by satisfactory repeatability and intermediate precision metrics. Nevertheless, certain limitations warrant consideration. The observation of negative relative bias values at lower concentrations suggests a potential underestimation or even non-detection of analytes in this range. This implies that the actual concentrations of leachable diisocyanates in wound dressings may even be higher than those reported in this study. As false positives are unlikely, the sensitisation risk for the leachable diisocyanates could even be greater than anticipated.

5 | Conclusion

A novel analytical method was developed to assess the leaching of residual diisocyanate monomers from medical devices, specifically wound dressings, under physiologically relevant

TABLE 6 | Risk characterisation for skin sensitisation induction and triggering allergic reactions in already sensitised individuals due to diisocyanates leaching from wound dressings.

Sample	Allergen	AEL	ANEAD	CEL	AEL/CEL	ANEAD/CEL	Sample	Allergen	AEL	ANEAD	CEL	AEL/CEL	ANEAD/CEL
Sample 1	HDI	0.0042	0.004	0.0027	1.5556	1.4815	Sample 9	DMDI	0.0500	0.009	0.0002	250	45
	DMDI	0.0500	0.009	0.0001	500	90		MDI	0.0167	0.007	0.0002	83.5000	35
	MDI	0.0167	0.007	0.0163	1.0245	0.4294		HDI	0.0042	0.004	0.0462	0.0909	0.0866
Sample 2	HDI	0.0042	0.004	0.0241	0.1743	0.1660	Sample 10	TDI 2,4	0.0167	0.005	0.0039	4.2821	1.2821
	TDI 2,4	0.0167	0.005	0.0018	9.2778	2.7778		DMDI	0.0500	0.009	0.0054	9.2593	1.6667
	DMDI	0.0500	0.009	0.0014	35.7143	6.4286		MDI	0.0167	0.007	0.0431	0.3875	0.1624
Sample 3	MDI	0.0167	0.007	0.0846	0.1974	0.0827	Sample 11	HDI	0.0042	0.004	0.0012	3.5000	3.3333
	HDI	0.0042	0.004	0.0079	0.5316	0.5063		DMDI	0.0500	0.009	0	NA	NA
	TDI 2,4	0.0167	0.005	0.0022	7.5909	2.2727		MDI	0.0167	0.007	0.0023	7.2609	3.0435
Sample 4	DMDI	0.0500	0.009	0.0008	62.5000	11.2500	Sample 12	HDI	0.0042	0.004	0.0098	0.4286	0.4082
	MDI	0.0167	0.007	0.0016	10.4375	4.3750		DMDI	0.0500	0.009	0.0004	125	22.5000
	HDI	0.0042	0.004	0.0637	0.0659	0.0628		MDI	0.0167	0.007	0.0002	83.5000	35
Sample 5	TDI 2,4	0.0167	0.005	0.0055	3.0364	0.9091	Sample 13	HDI	0.0042	0.004	0.0047	0.8936	0.8511
	DMDI	0.0500	0.009	0.0033	15.1515	2.7273		TDI 2,4	0.0167	0.005	0.0047	3.5532	1.0638
	MDI	0.0167	0.007	0.0043	3.8837	1.6279		DMDI	0.0500	0.009	0.0006	83.3333	15
Sample 6	HDI	0.0042	0.004	0.0181	0.232	0.221	Sample 14	MDI	0.0167	0.007	0.0008	20.8750	8.7500
	DMDI	0.0500	0.009	0.0001	500	90		HDI	0.0042	0.004	0.0006	7	6.6667
	MDI	0.0167	0.007	0.0009	18.5556	7.7778		TDI 2,4	0.0167	0.005	0.0021	7.9524	2.3809
Sample 7	HDI	0.0042	0.004	0.0154	0.2727	0.2597	Sample 15	MDI	0.0167	0.007	0.0001	167	70
	DMDI	0.0500	0.009	0.0027	18.5185	3.3333		HDI	0.0042	0.004	0.0033	1.2727	1.2121
	MDI	0.0167	0.007	0.0044	3.7955	1.5909		DMDI	0.0500	0.009	0.0006	83.3300	15
Sample 8	HDI	0.0042	0.007	0.0082	0.5122	0.8537	Sample 16	MDI	0.0167	0.007	0.002	8.3500	3.5000
	TDI 2,4	0.0167	0.005	0.0011	15.1818	4.5455		HDI	0.0042	0.004	0.0023	1.8261	1.7391
	DMDI	0.0500	0.009	0.0003	166.6660	30		MDI	0.0167	0.007	0.0023	1.8261	1.7391
Sample 9	MDI	0.0167	0.007	0.0485	0.3443	0.1443	Sample 17	MDI	0.0167	0.007	0.0023	1.8261	1.7391
	HDI	0.0042	0.004	0.0048	0.8750	0.8333		MDI	0.0167	0.007	0.0023	1.8261	1.7391
	DMDI	0.0500	0.009	0.001	50	9		MDI	0.0167	0.007	0.0023	1.8261	1.7391
Sample 10	MDI	0.0167	0.007	0.0062	2.6935	1.1290	Sample 18	MDI	0.0167	0.007	0.0023	1.8261	1.7391
	HDI	0.0042	0.004	0.0062	2.6935	1.1290		MDI	0.0167	0.007	0.0023	1.8261	1.7391
	DMDI	0.0500	0.009	0.0062	2.6935	1.1290		MDI	0.0167	0.007	0.0023	1.8261	1.7391

Note: Values in bold indicate cases where the consumer exposure level (CEL) ($\mu\text{g}/\text{cm}^2$) exceeds the acceptable non-eliciting area dose (ANEAD) ($\mu\text{g}/\text{cm}^2$) and the acceptable exposure level (AEL) ($\mu\text{g}/\text{cm}^2$), suggesting a potential risk to consumers both in sensitisation induction and allergy elicitation for consumers already sensitised.

Abbreviations: DMDI, Dicyclohexylmethane-4,4-diisocyanate; HDI, Hexamethylene diisocyanate; MDI, 4,4'-diphenylmethane diisocyanate; NA, Non-attributed; TDI 2,4, Toluene-2,4-diisocyanate.

Graphical representation of the Consumer Exposure Level (CEL)

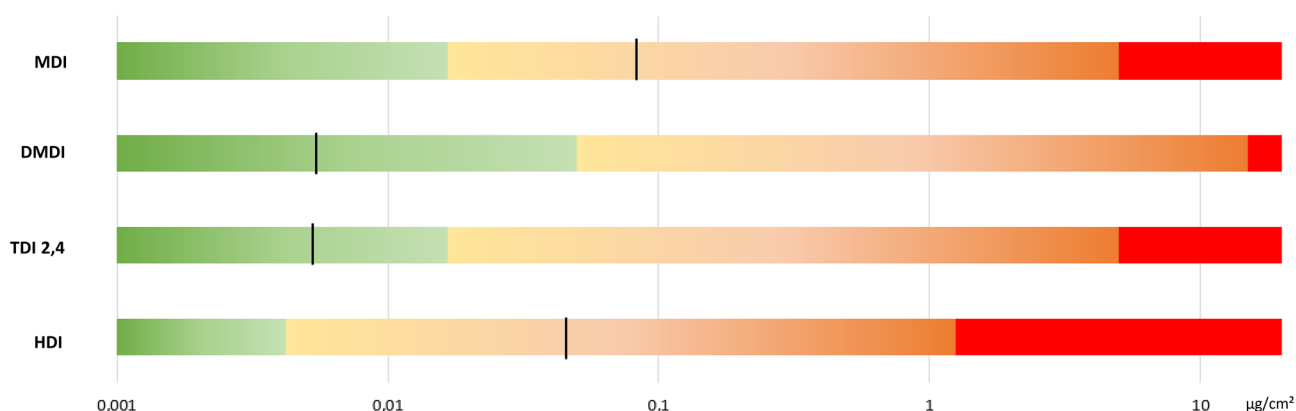


FIGURE 4 | Graphical representation of the worst cases of Consumer Exposure Level (CEL) determined from the assessed samples (adapted from Marcelis et al. [50]). Green: Below the Acceptable Exposure Limit (AEL); Orange: Within the Safety Assessment Factors (SAFs) range and Red: Above the No Expected sensitisation Induction Level (NESIL). CEL values indicated (black lines) are for MDI: 0,0846 (sample 2); DMDI: 0,0054 (sample 10); TDI 2,4: 0,0055 (sample 4), and HDI: 0,0462 (sample 10). For both HDI and MDI, the CELs fall within the range of the SAFs. Both TDI-2,4 and DMDI present CELs below the AEL. DMDI, Dicyclohexylmethane-4,4'-diisocyanate; HDI, Hexamethylene diisocyanate; MDI, 4,4'-diphenylmethane diisocyanate; TDI 2,4, Toluene-2,4-diisocyanate.

conditions and simulating realistic skin contact. This approach overcomes limitations of previous methods by better reflecting actual exposure scenarios and potential interactions with skin proteins that could trigger sensitisation.

The detection of high concentrations of potent sensitizers such as HDI is particularly concerning given the prolonged application of wound dressings to compromised skin, which facilitates chemical absorption and increases sensitisation induction risk, even at low doses. Additionally, individuals already sensitised may face higher risks of developing ACD, especially due to known diisocyanate cross-reactivity. Overall, these findings underscore the need for more rigorous safety evaluations and regulatory oversight of residual diisocyanates in medical devices.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author, B.D., upon reasonable request.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Data S1:** Supporting Information. **Data S2:** Supporting Information.